

# Solving Surpassing the Leader

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With optimal play, Hal wins about **51.86%** of games from the 8:12 opening. We compute **9.45 billion reachable state classes** with the leap second available to Baku.

## 1. From DTH to STL

We extend *Solving Drop the Handkerchief* [1]. We keep its rules and survival model, with  $s$  for squandered time (ST) and  $t$  for total time dead (TTD). Both players start with  $(s, t) = (0, 0)$  at 8:12 AM. Hal drops first and Baku checks.

Timekeepers add a leap second to keep atomic-clock time aligned with Earth’s rotation. At a scheduled year-end insertion, the UTC clock passes through 23:59:60 on December 31. Japan is nine hours ahead, so the extra second occurs at 8:59:60 AM on January 1 [2].

The match takes place on January 1, with a leap second due that morning. Yakou uses a broadcast speaking clock. Baku plans to be the Dropper at 8:59, so he can hold the handkerchief through 60 seconds and drop during the extra second, before the clock reaches 9:00. He still drops within that clock minute [3].

We model this by allowing Baku as Dropper to choose second 61 in that minute. The Checker remains capped at 60, so Baku can force a failed check. Both players know this rule from the start in our solve. They must account for who will drop in the leap window and how much time an injection will consume before the next turn.

We add player identity and a clock to the DTH state:

$$x = (s_c, t_c, s_d, t_d, h, \tau).$$

Hal drops in half  $h = 1$ ; Baku drops in half  $h = 2$ . We measure  $\tau$  in physical seconds from 8:00, including the leap second. The opening has  $h = 1$  and  $\tau = 720$ .

We use the 17,011 player profiles from the DTH quotient [1, Section 2]. At a fixed half and clock, the same grouping holds: a successful check does not use TTD, and a failed check ends the game for a failure-fatal profile. We count 9,453,333,117 reachable classes across 2,243 half-and-clock groups. After Baku revives, we need only 183 Dropper profiles because his ST is zero.

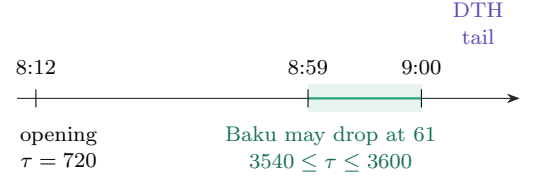
## 2. The clock

Let  $T(\tau) = 61$  for  $3540 \leq \tau \leq 3600$ , and 60 otherwise. We set  $q = 0$  after success and use the injected dose  $q = s_c + 60$  after revival. For a live child, define

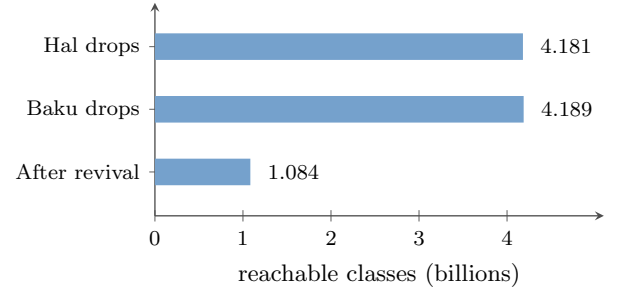
$$e = \tau + T(\tau) + (q + 120)\mathbf{1}_{q>0}, \quad \tau' = \begin{cases} e + 60, & h = 1, \\ \nu(e), & h = 2. \end{cases}$$

Here  $\nu(e)$  is the next minute boundary strictly after  $e$ , with 9:00 at  $\tau = 3601$ . We swap roles and change  $h$ . The dose therefore affects when the next player can act.

We solve clock groups in reverse order, with each live child before its parent. For  $\tau' > 3600$ , we use the DTH solution: both players have the base game’s action sets for the rest of play.



**Figure 1:** Baku can drop at 61 in the leap window. After 9:00, both players use seconds 1 to 60. The timeline is schematic.



**Figure 2:** We count each reachable quotient state once. The revival group contains Baku’s turns after he survives an injection.

## 3. The extra second

We use the DTH payoff matrix  $M$  [1, Section 4], with success payoffs  $S_\ell$  and failure payoff  $F$ . The value  $V$  is the current Dropper’s win probability minus loss probability; a role swap negates it. We evaluate the continuation states at their new clocks. Successes share one child clock; failures share another for the given Checker profile. We still need only 60 success payoffs and  $F$ . In Baku’s window, we add a row of 60 copies of  $F$ . If  $v_{60}$  is the square matrix value, then

$$v_{61} = \max\{v_{60}, F\}.$$

**Proof.** Give the extra row mass  $\alpha$  and an ordinary Dropper mixture  $a$  the remaining mass. The guaranteed payoff is  $\alpha F + (1 - \alpha) \min_c(a^\top M)_c$ . We maximize over  $a$ , then choose the better endpoint in  $\alpha$ . The Checker can use a minimizing mixture for  $M$  in either case. QED

#### 4. Solving the round

We use the fixed-action test and mixed-strategy recurrence from DTH [1, Section 4]. We check each candidate against the same  $10^{-6}$  saddle-gap limit. The extra second changes the continuation values, and some states need linear programming to meet this limit.

#### 5. Linear programming

For an unresolved stage with  $\delta = F - S_1 > 0$ , define

$$A_{ij} = \begin{cases} (F - S_{j-i+1})/\delta, & j \geq i, \\ 0, & j < i. \end{cases}$$

Then  $M = F\mathbf{1}\mathbf{1}^\top - \delta A$ , with unit diagonal in  $A$ . We solve the dual linear programs

$$\max_{y \geq 0} \{\mathbf{1}^\top y : A^\top y \leq \mathbf{1}\} = \min_{z \geq 0} \{\mathbf{1}^\top z : Az \geq \mathbf{1}\} = Z.$$

**Proof.** The unit diagonal gives a finite nonnegative covering vector by back substitution; zero is feasible for packing. Linear-programming duality gives a common optimum  $Z > 0$ . For  $a = y/Z$  and  $b = z/Z$ , we obtain  $a^\top M \geq (F - \delta/Z)\mathbf{1}^\top$  and  $Mb \leq (F - \delta/Z)\mathbf{1}$ . Thus  $v_{60} = F - \delta/Z$ . QED

We solve the remaining stages by linear programming. For a candidate Dropper mixture  $a$  and Checker mixture  $b$ , we bound the value with

$$L = \min_c (a^\top M)_c, \quad U = \max_d (Mb)_d.$$

In the leap window, we replace each bound by its maximum with  $F$ . We require  $U - L \leq 10^{-6}$  and use the midpoint or a center with proved error at most  $5 \times 10^{-7}$ .

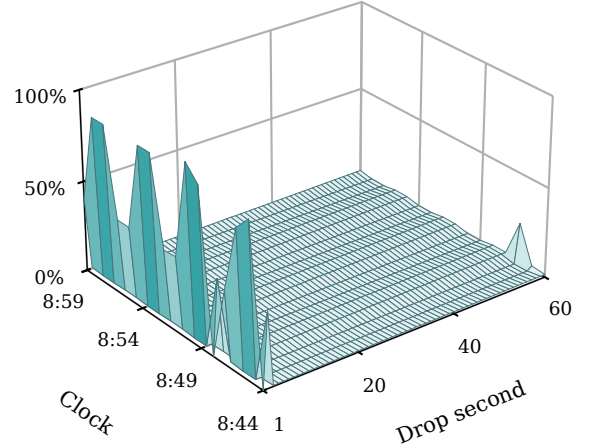
#### 6. Error bounds

We retain the DTH progress score  $\Phi$  and its bound of 1,201 steps [1, Sections 3 and 5]. The clock adds no cycles, and the extra failure row uses the same injection transition. Thus the DTH error proof also covers this extension:

$$|\hat{V} - V| \leq (1201 - \Phi)5 \times 10^{-7}.$$

At the opening, the win-probability error is at most 0.000301. We use floating-point saddle checks for the matrix and LP paths. We re-solve 4,621 sampled stages by linear programming. The largest difference from the stored values is  $3.07 \times 10^{-9}$ .

Hal's drop probability



**Figure 3:** Hal's drop strategy at 16 reachable states, one per minute from 8:44 to 8:59. Hal has  $(s, t) = (0, 180)$ ; Baku has  $(60, 120)$ . We hold these profiles fixed and join adjacent clock slices. Each slice is a separate state. Hal puts 95.77% on second 1 at 8:50 and 41.31% at 8:51.

#### 7. Results

At the starting state, we obtain

$$V = 0.03717 \pm 0.00061, \\ P(\text{Hal wins}) = 0.51859 \pm 0.00031.$$

We solve 8,843,634,465 classes with fixed actions or the recurrence and gap check, and 609,698,652 with linear programming.

Baku wins 48.14% of games, a gain of 2.63 percentage points over the base game. Hal's win probability falls from 54.49% to 51.86%. A successful check and a survived injection lead to different clocks. The players must account for who will drop when the extra second arrives, so their strategies can change before the leap window even at fixed loads (Figure 3).

**References.** [1] J. Cenawood, *Solving Drop the Handkerchief*, September 11, 2026. [2] NICT, *Japan Standard Time; leap-second notice*, October 2008. [3] T. Sako, *Usogui*, Shueisha, 2006–2018.